Artificial Gravity

Answer key

Conventions used throughout

• **Apparent weight** = the normal force F_N (2.6). “0.4g” means $F_N = 0.4mg$.

• **Rotating spacecraft:** in deep space the floor (the outer wall) is the only thing pushing on the astronaut, so F_N is the whole centripetal force: $F_N = ma_c$, with $a_c = v^2/r = 4\pi^2 rf^2$. For “kg’s”, $a_c = k \times 9.81$ m/s^2.

• **Units:** f in Hz (rotations per second); rpm = f × 60. T = 1/f.

• **Constants:** g = 9.81 m/s^2, G = 6.67 × 10^{-11} N·m^2/kg^2, m_E = 5.98 × 10^{24} kg, r_E = 6.38 × 10^6 m. Every step keeps full precision; only final answers are rounded.

HERMES

Artificial gravity by rotation

As you know, the sensation of weight, does not come from the force of gravity, rather, it comes from the _____ force. One way to produce artificial gravity is to have the spacecraft **rotate**.

In the movie, “The Martian”, the *Hermes* spacecraft rotated to produce artificial gravity.

a) Where would an astronaut be standing to get the effects of artificial gravity? Draw this on the picture above. Check with the teacher

b) Based on your diagram, draw a free body diagram for the astronaut at this point.

c) In the novel, *The Martian*, the spacecraft creates 0.4g’s (i.e. the astronauts experience 0.4 times their regular weight when on the ship.) What does this tell you about their normal force?

d) Now estimate the radius of the *Hermes*. Say the pods at the edge of the circle are 3.0 m high.

e) Use your estimate and FBD to calculate the required frequency for the *Hermes* to create the required artificial gravity

f) If the *Hermes* needed to create 1g of weight (rather than 0.4g) what frequency of rotation would it need? Convert your answer to rpm’s (rotations per minute) and discuss how an astronaut might feel on this ship.

g) Now say NASA wanted to reduce materials costs and created a rotating spacecraft with a radius of only 2.0 m. With what frequency would it have to rotate to create 0.4g? What about 1g?

h) Convert your answers in (g) to rpm’s and discuss the practicality of this design.

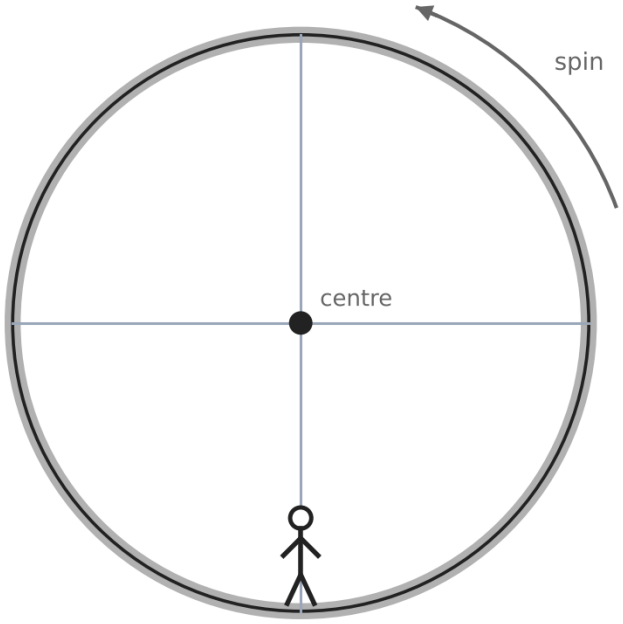
Blank the sensation of weight comes from the _____ force

normal force (2.6: a scale, and your body, feel the push of the floor, not gravity itself).

a) Where would an astronaut be standing to get the effects of artificial gravity?

On the inside of the outer wall of the spinning ring (in the pods at the edge of the circle), with feet on the outer wall and head toward the centre. The wall pushes the astronaut toward the centre, which keeps them moving in a circle, and that push feels like a floor.

Where the astronaut stands (end view of the ring)



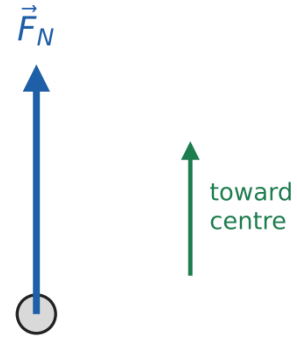
astronaut stands on the outer wall, head toward the centre

b) Draw a free body diagram for the astronaut at this point.

Far from planets, gravity on the astronaut is tiny, so the only force is the normal force from the wall, pointing toward the centre.

$$\Sigma F_c = ma_c \quad \Rightarrow \quad F_N = ma_c$$

FBD of the astronaut



no gravity drawn (deep space)

c) What does this tell you about their normal force?

Feeling 0.4 times their weight means the wall pushes with 0.4 of their Earth weight. So $F_N = ma_c = 0.4mg$, which means $a_c = 0.4g$:

$$F_N = 0.4mg \quad \Rightarrow \quad a_c = 0.4(9.81) = 3.924 \text{ m/s}^2$$

d) Now estimate the radius of the *Hermes*.

In the photo the ring's radius (centre to the outer edge of the pods) is about 4 times the height of a pod. With 3.0 m pods:

$$r \approx 4 \times 3.0 \text{ m} = 12 \text{ m}$$

This is an estimate from a picture. Your own estimate may differ a little; use it in (e) and (f) the same way.

e) Calculate the required frequency for the *Hermes* to create the required artificial gravity

Solve $a_c = 4\pi^2 r f^2$ for f:

Formula: $a_c = 4\pi^2 r f^2$

Divide by $4\pi^2 r$ and square root: $f = \sqrt{\dfrac{a_c}{4\pi^2 r}}$

$$f = \sqrt{\frac{3.924}{4\pi^2(12)}} = 0.0910 \text{ Hz}$$

f) What frequency for 1g? Convert to rpm and discuss.

$$f = \sqrt{\frac{9.81}{4\pi^2(12)}} = 0.1439 \text{ Hz}$$

$$\text{rpm} = (0.1439)(60) = 8.6 \text{ rpm}$$

About 9 turns a minute. The astronaut would feel normal weight, but turning the head could cause dizziness and motion sickness, and the “gravity” changes noticeably between feet and head, because a_c depends on r. (At 0.4g it is 5.5 rpm.)

g) A radius of only 2.0 m: what frequency for 0.4g? For 1g?

$$0.4g: \quad f = \sqrt{\frac{3.924}{4\pi^2(2.0)}} = 0.2229 \text{ Hz}$$

$$1g: \quad f = \sqrt{\frac{9.81}{4\pi^2(2.0)}} = 0.3525 \text{ Hz}$$

h) Convert your answers in (g) to rpm's and discuss the practicality of this design.

$$0.4g: \ (0.2229)(60) = 13.4 \text{ rpm} \qquad 1g: \ (0.3525)(60) = 21.1 \text{ rpm}$$

Not practical. Spinning 13 to 21 times a minute would make astronauts very dizzy and sick. Also, a 2.0 m radius is about the height of a person, so the head would be near the centre and feel almost no gravity while the feet feel the full amount.

FINAL ANSWER

Blank: normal

a) On the inside of the outer wall of the ring, head toward the centre

b) One force: F_N toward the centre

c) $F_N = 0.4mg$, so $a_c = 0.4g$

d) r ≈ 12 m (estimate)

e) f ≈ 0.091 Hz

f) f ≈ 0.14 Hz ≈ 8.6 rpm; likely dizziness

g) 0.4g: 0.22 Hz; 1g: 0.35 Hz

h) 13 rpm and 21 rpm: far too fast to be comfortable

ISS

Are astronauts orbiting the Earth weightless?

A 57 kg astronaut is on the International Space Station which is orbiting 660 km above the surface of the earth. Weight is DEFINED as the force of gravity acting on an object.

a) Find the weight of the astronaut. [459 N]

b) What is the astronaut's weight on the earth? [559 N]

c) Is the astronaut weightless? If not, what percentage of her earth weight does she have in the space station? [~82%]

d) Why does the astronaut APPEAR to be weightless even though she clearly has weight?

a)

Use universal gravitation with the centre-to-centre distance r = $6.38 \times 10^6 + 6.6 \times 10^5 = 7.04 \times 10^6$ m:

$$F_g = \frac{Gm_E m}{r^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(57)}{(7.04 \times 10^6)^2} = 458.7 \text{ N}$$

b)

$$F_g = mg = (57)(9.81) = 559.2 \text{ N}$$

c)

No. Gravity still pulls on her with:

$$\frac{458.7}{559.2} \times 100\% = 82.0\%$$

d)

She and the station are both in free fall around Earth (2.10). They fall together, so the floor doesn't push on her: $F_N = 0$. We feel weight through the normal force, so with no normal force she feels weightless, even though gravity is about 82% as strong as on the ground.

FINAL ANSWER

a) $F_g = 4.6 \times 10^2$ N

b) $F_g = 5.6 \times 10^2$ N

c) No: about 82% of her Earth weight

d) She falls freely with the station, so $F_N = 0$ and she feels no weight

Worksheet answer: a) 459 N; b) 559 N
Why the answers differ: Same values, different rounding. The unrounded weights are 458.7 N and 559.2 N. The mass 57 kg has only 2 significant figures, so the key gives 4.6×10^2 N and 5.6×10^2 N; the worksheet keeps 3 figures. Part (c) is unaffected: 458.7 ÷ 559.2 = 82%.

CLASS WORK 1

Large rotating space station

1. You are an astronaut on a rotating space station. Your station has an inside diameter of 3.0 km

   a. Determine the speed required to create 1g of artificial gravity. Include an FBD. [121 m/s]

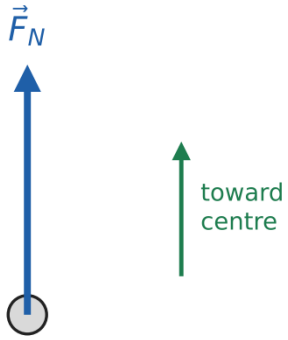
   b. Determine the period of rotation of the spacecraft. [78 s]

r = 3.0 km ÷ 2 = 1500 m. As on the *Hermes*, the floor's normal force is the only force and points to the centre (FBD below). For 1g, $F_N = mg$:

a. $$F_N = \frac{mv^2}{r} = mg \quad \Rightarrow \quad v = \sqrt{gr} = \sqrt{(9.81)(1500)} = 121.31 \text{ m/s}$$

b. $$T = \frac{2\pi r}{v} = \frac{2\pi(1500)}{121.31} = 77.7 \text{ s}$$

FBD of the astronaut



no gravity drawn (deep space)

FINAL ANSWER

a. $v = 1.2 \times 10^2$ m/s

b. T = 78 s

Worksheet answer: a. 121 m/s
Why the answers differ: Same value, different rounding. √((9.81)(1500)) = 121.3 m/s. The diameter 3.0 km has 2 significant figures, so the key gives 1.2×10^2 m/s; the worksheet keeps 3 figures (but gives (b), 78 s, to 2).

CLASS WORK 2

Spinning Skylab

2. The radius of the Skylab module used by NASA is about 6.6m. If they were to spin the module at 9.7 rpm, what fraction of their regular weight would a person experience? [70%]

Change rpm to Hz, find a_c, then compare it with g ($F_N = ma_c$, Earth weight = mg):

$$f = \frac{9.7}{60} = 0.1617 \text{ Hz}$$

$$a_c = 4\pi^2 r f^2 = 4\pi^2 (6.6)(0.1617)^2 = 6.810 \text{ m/s}^2$$

$$\frac{F_N}{mg} = \frac{a_c}{g} = \frac{6.810}{9.81} = 0.694$$

FINAL ANSWER

About 69% of their regular weight

Worksheet answer: 70%
Why the answers differ: The unrounded fraction is 0.694, or 69.4%. The data (6.6 m and 9.7 rpm) have 2 significant figures, so the answer is 69%. The worksheet's 70% rounds 69.4% to only 1 figure, which is less precise than the data allow.

CLASS WORK 3

Human-powered centrifuge

3. In a bicycle centrifuge with a 4.0 m diameter, an astronaut is pedalling while laying down as shown in the diagram below. They create enough force to equal 0.85 g's.

   a. Determine the speed, frequency and period of the centrifuge. [4.1 m/s; 0.32 Hz; 3.1 s]

   b. Does the speed you calculated in (a) apply to the astronaut's whole body, head or feet? Explain. What about the frequency and period?

   c. What might be a drawback to this design?

   d. How would the frequency change if the diameter was increased to 6.0 m? Would this make the astronaut feel more dizzy or less dizzy?

a. Determine the speed, frequency and period of the centrifuge.

r = 4.0 m ÷ 2 = 2.0 m (at the feet, on the outer edge). $a_c = 0.85(9.81) = 8.3385\ m/s^2$:

$$v = \sqrt{a_c r} = \sqrt{(8.3385)(2.0)} = 4.084\ \text{m/s}$$

$$f = \frac{v}{2\pi r} = \frac{4.084}{2\pi(2.0)} = 0.3250\ \text{Hz} \qquad T = \frac{1}{f} = 3.08\ \text{s}$$

b. Does the speed apply to the whole body, head or feet? What about the frequency and period?

Only to the feet, at the 2.0 m edge. Every part of the body goes around once in the same time, but the head is closer to the centre, so it travels a smaller circle and moves more slowly (v = 2πrf). The frequency and period are the same for the whole body.

c. What might be a drawback to this design?

Because $a_c = 4\pi^2 r f^2$ depends on r, the feet feel 0.85g but the head, much closer to the centre, feels far less. That big difference along the body, plus spinning about 19 times a minute, can cause dizziness and nausea. (It also takes a lot of pedalling effort.)

d. How would the frequency change if the diameter was increased to 6.0 m?

r = 3.0 m, same a_c:

$$f = \sqrt{\frac{a_c}{4\pi^2 r}} = \sqrt{\frac{8.3385}{4\pi^2(3.0)}} = 0.2653 \text{ Hz}$$

The frequency drops (from 0.32 Hz to 0.27 Hz). Spinning more slowly makes the astronaut feel less dizzy.

FINAL ANSWER

a. v = 4.1 m/s, f = 0.32 Hz, T = 3.1 s

b. The speed is for the feet only; the frequency and period are the same for the whole body

c. The head feels much less “gravity” than the feet, and the fast spin causes dizziness

d. f decreases to 0.27 Hz, so the astronaut feels less dizzy

POPULAR MECHANICS

Bigger radius, slower spin

A Popular Mechanics Article quoted Professor John Page: "The smaller the spacecraft is, the faster it has to rotate," he says, "so if you're going to generate gravity, it's got to be done with a very large spacecraft that spins very slowly. The bigger the disk, the slower you can rotate it." Do a calculation to show that a larger radius decreases the necessary frequency.

Step 1 – The formula

For artificial gravity of 1g, the wall must give $a_c = g$. Solve for f:

$$g = 4\pi^2 r f^2 \quad \Rightarrow \quad f = \sqrt{\frac{g}{4\pi^2 r}}$$

r is in the denominator, so a larger r gives a smaller f.

Step 2 – Numbers for 1g

$$r = 2.0\ \text{m}: \quad f = 0.352\ \text{Hz} = 21.1\ \text{rpm}$$

$$r = 12\ \text{m}: \quad f = 0.144\ \text{Hz} = 8.6\ \text{rpm}$$

$$r = 1500\ \text{m}: \quad f = 0.0129\ \text{Hz} = 0.77\ \text{rpm}$$

Each larger radius needs a slower spin, exactly as Professor Page says. (Making r 4 times bigger halves f, because f ∝ 1/√r.)

FINAL ANSWER

f = √(g ÷ 4π^{2}r): 2.0 m needs 21 rpm, but 1500 m needs only 0.77 rpm